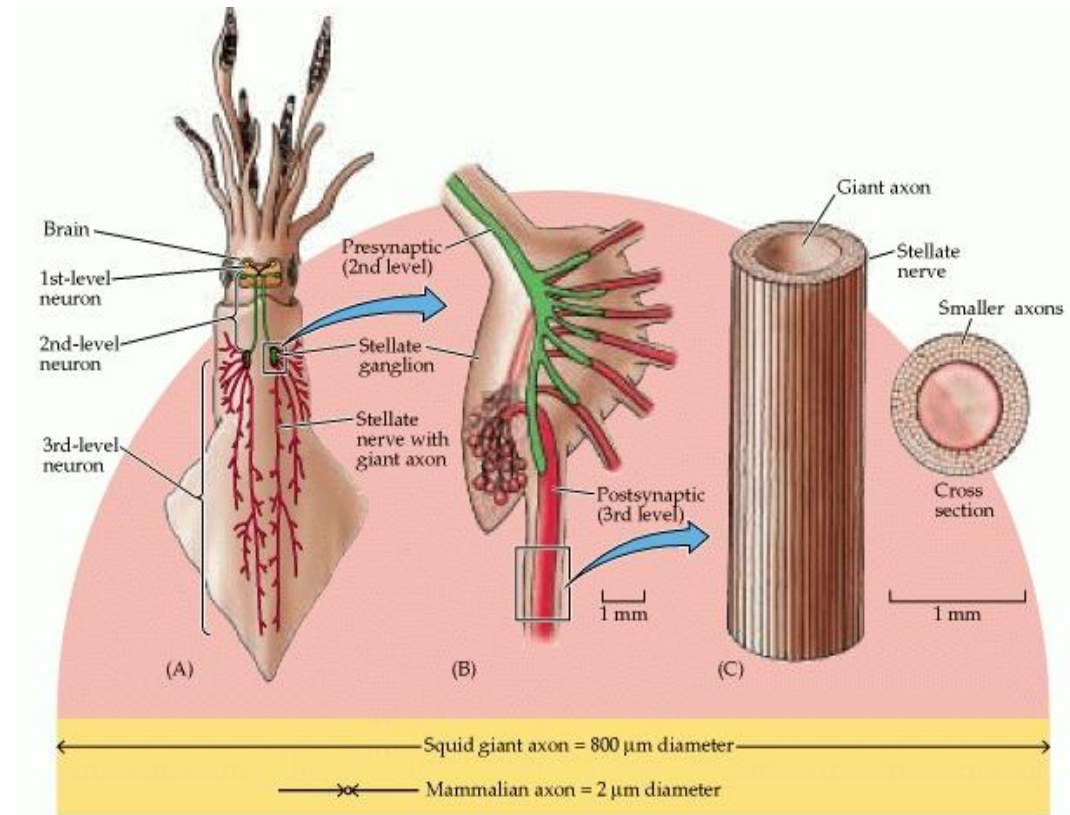


THE PATCH CLAMP TECHNIQUE

Fernando C. Ortiz, PhD
fernando.ortiz.c@usach.cl

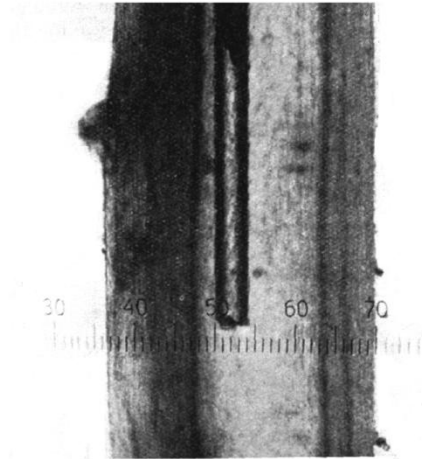
Mechanisms of Myelin Formation and Repair Laboratory
Departamento de Biología, Facultad de Química y Biología
Universidad de Santiago de Chile

Bioelectricity

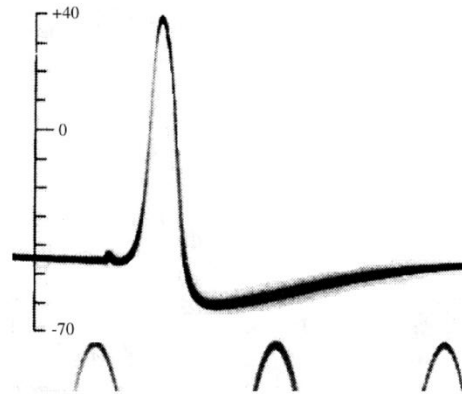


Bioelectricity: the squid giant axon

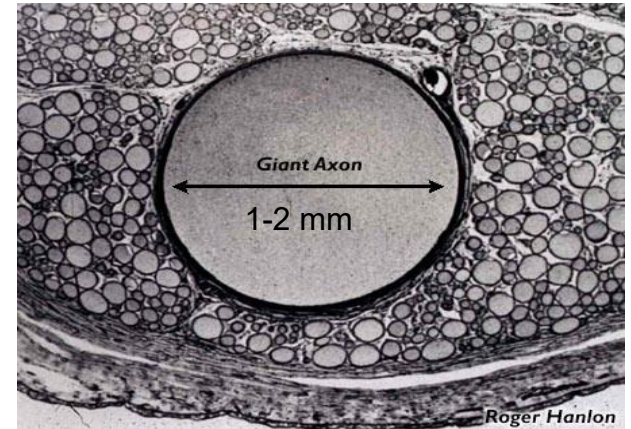
A



B

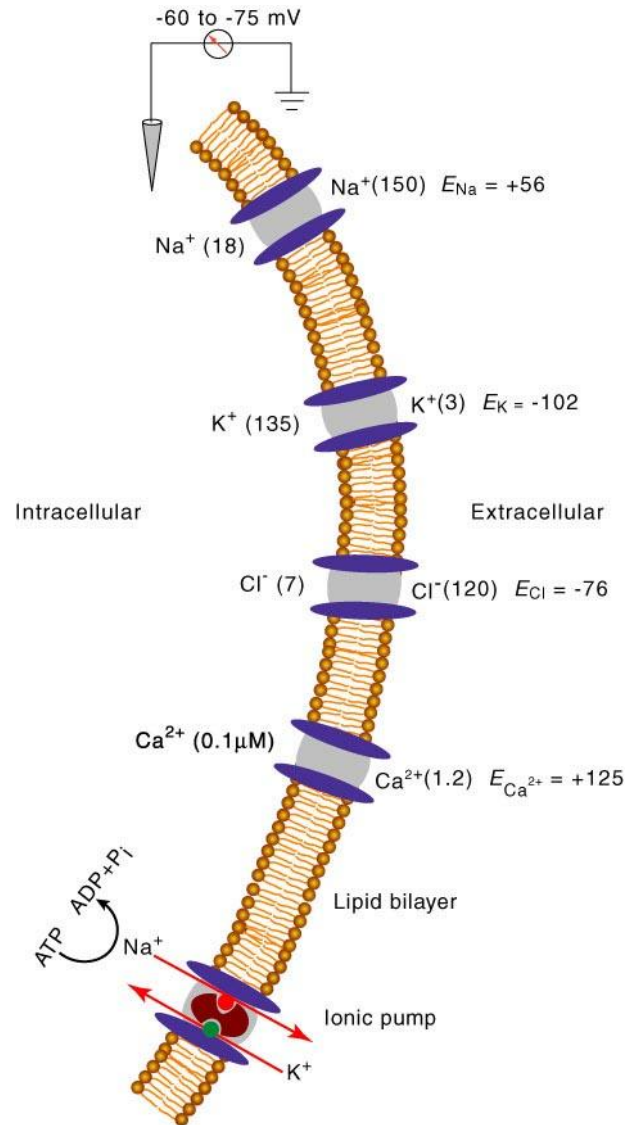


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- Hodgkin & Huxley (1939) (and Curtis & Cole (1940)) recorded the membrane potential directly from the squid giant axon
- Nobel prize 1963

The resting membrane potential (RMP or V_m)



Nernst:

$$E_c = \frac{RT}{zF} \ln (C_e/C_i)$$

one ion

All of them

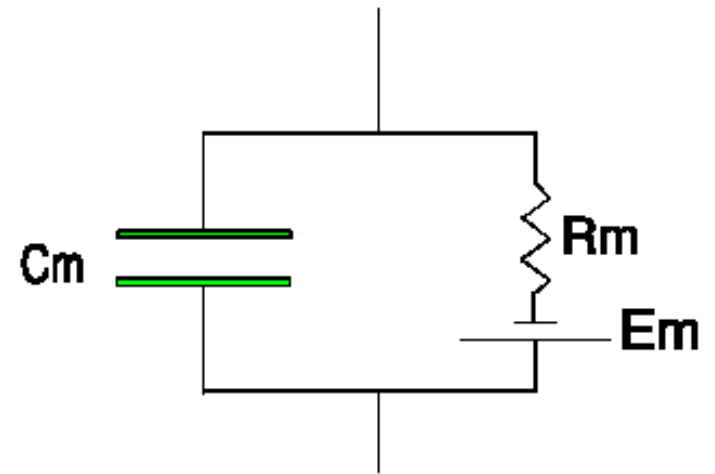
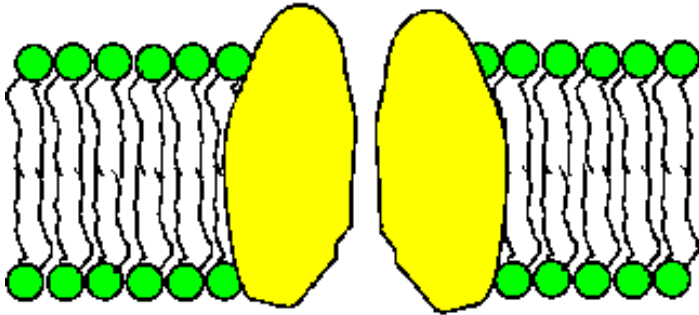
Goldman, Hodgkin & Katz eq.:

$$V_m \rightarrow \frac{RT}{zF} \ln (K_e/K_i) + \frac{RT}{zF} \ln (Na_e/Na_i) + \frac{RT}{zF} \ln (Cl_i/Cl_e)$$

$$V_m = \frac{RT}{zF} \ln \frac{(P_K [K]_o + P_{Na} [Na]_o + P_{Cl} [Cl]_i)}{(P_K [K]_i + P_{Na} [Na]_i + P_{Cl} [Cl]_o)}$$

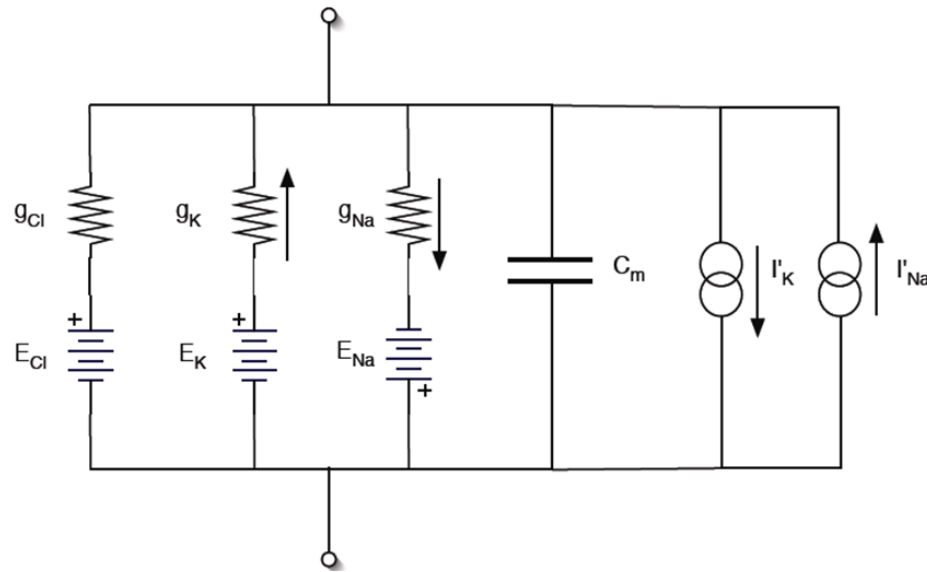
What about biological systems?

RC circuit



- membrane lipids as a capacitor(C): charge (Q) storage
- Ion channels as resistors (R)
- RMP as a Battery (E).

RC circuit

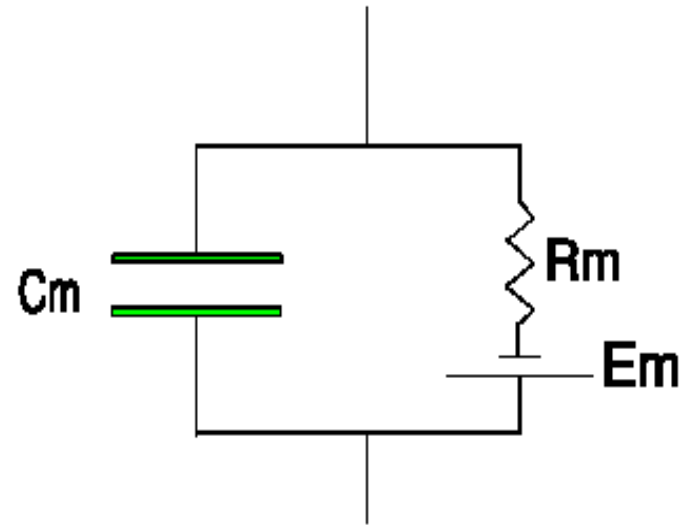
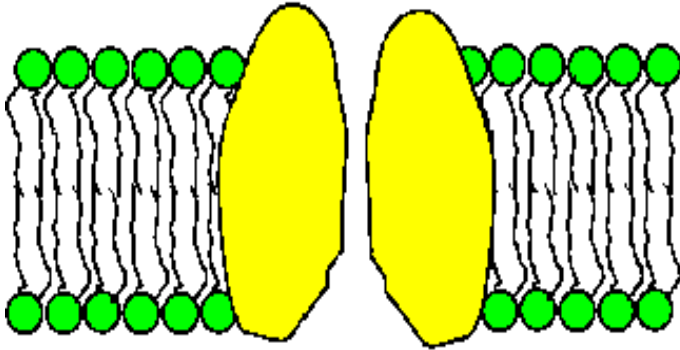


¿what about ion currents trough the membrane?

$$I_m = I_R + I_C$$

$$I_m = \frac{V_m}{R_m} + C_m \frac{dV_m}{dt}$$

RC circuit



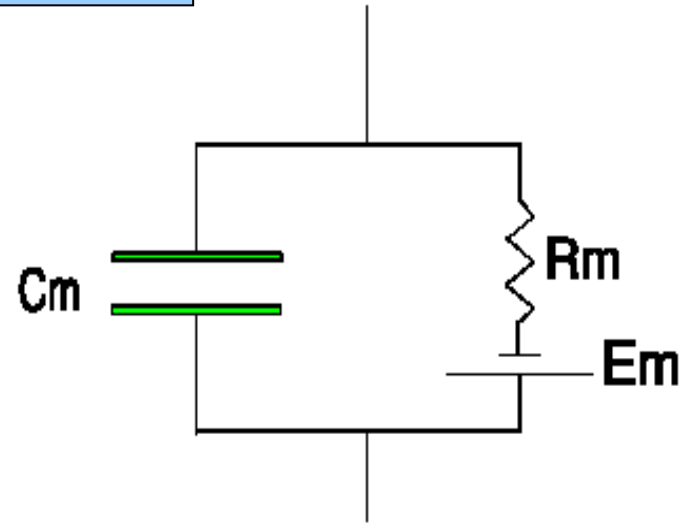
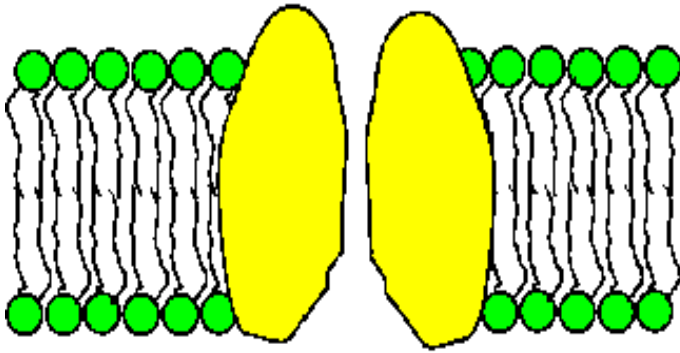
$$I_T = I_R + I_C$$

$$I_T = \frac{V}{R_m} + C_m \frac{dV_m}{dt}$$

$$I_C = C_m \frac{dV_m}{dt}$$

... then, when V is constant $\rightarrow I_T = I_R$

RC circuit



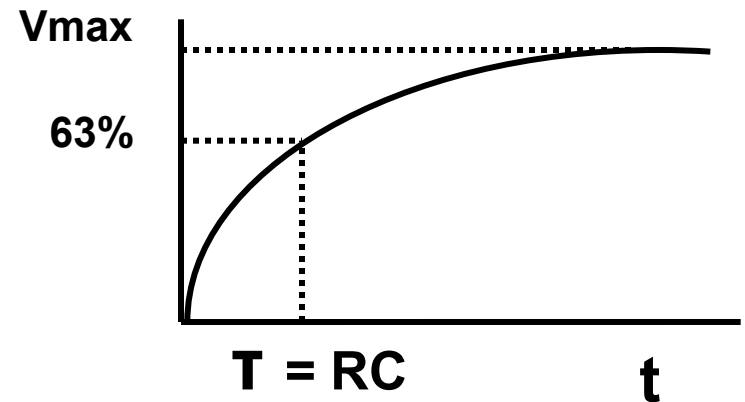
$$I_T = \frac{V}{R_m} + C_m \frac{dV_m}{dt}$$

$$V_{(t)} = I_o R_m (1 - e^{-t/R_m C_m}) \text{-charging}$$

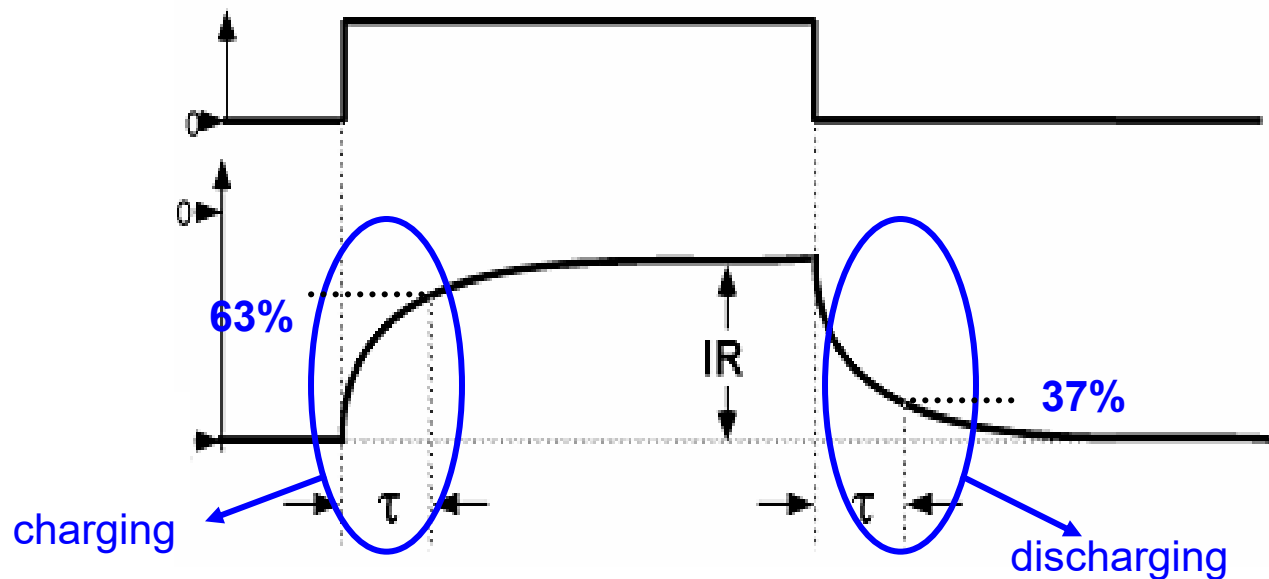
$$V_{(t)} = V_{\max} (1 - e^{-t/\tau}) \text{-discharging}$$

$$(\tau = R_m C_m)$$

τ = time constant



Capacitor charging and discharging

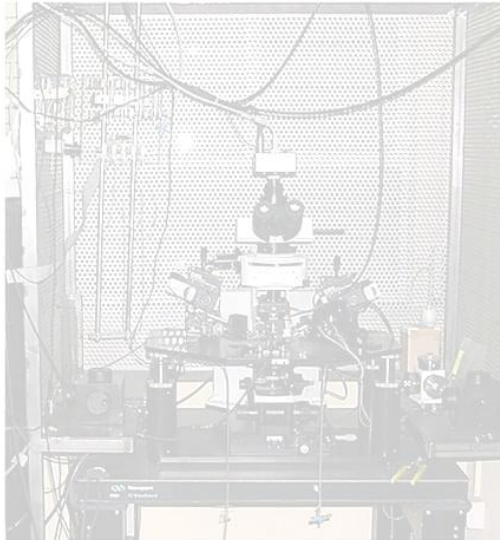


$$V_{(t)} = V_{\max}(1 - e^{-t/\tau})$$

$$V_{(t)} = V_{\max}e^{-t/\tau}$$

Potential does not change instantaneously in response to a current change
(time course is τ dependent)

A



B

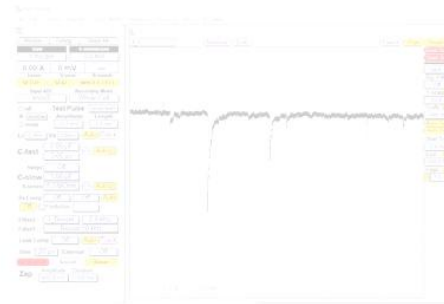


C

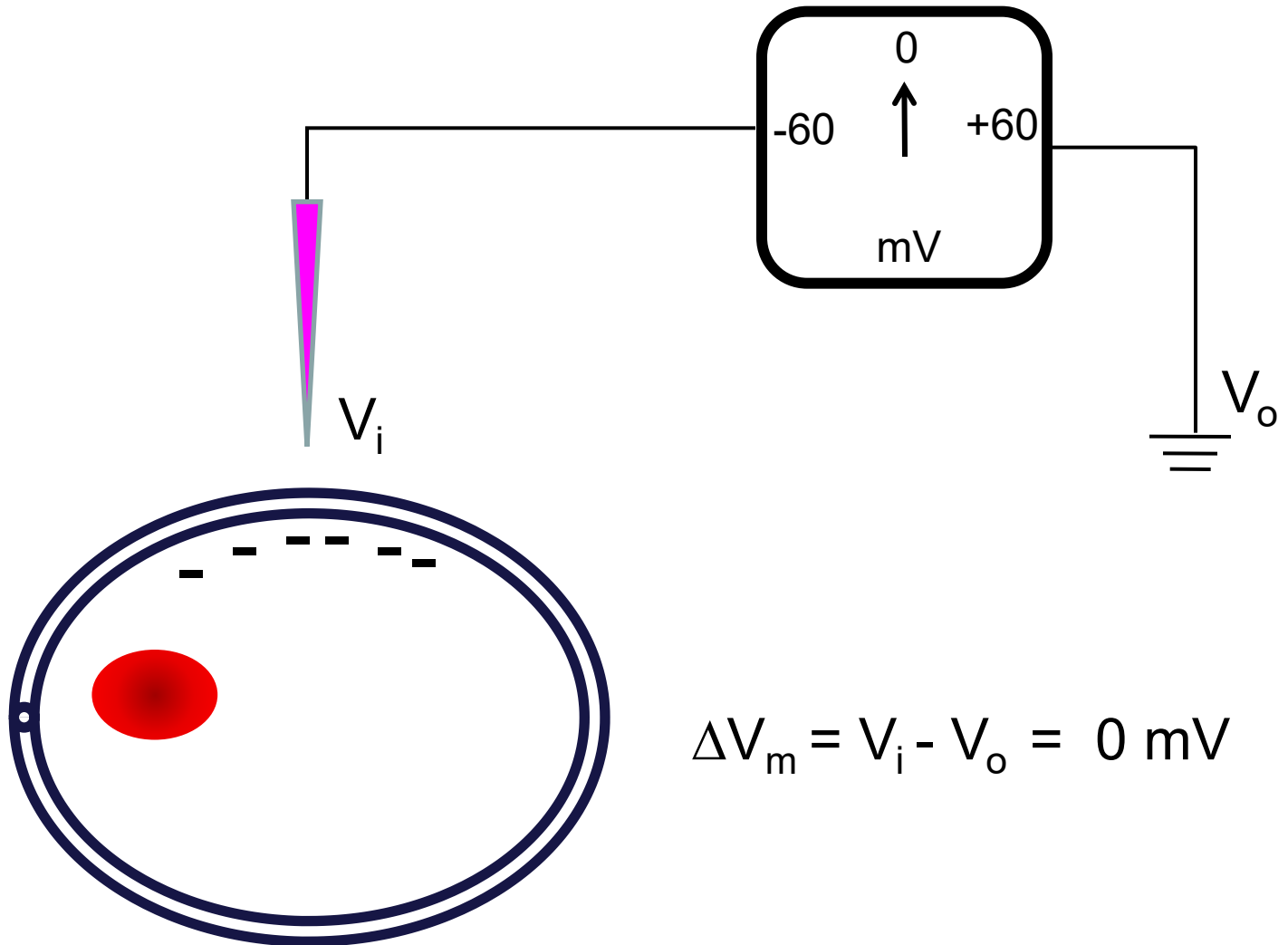


D

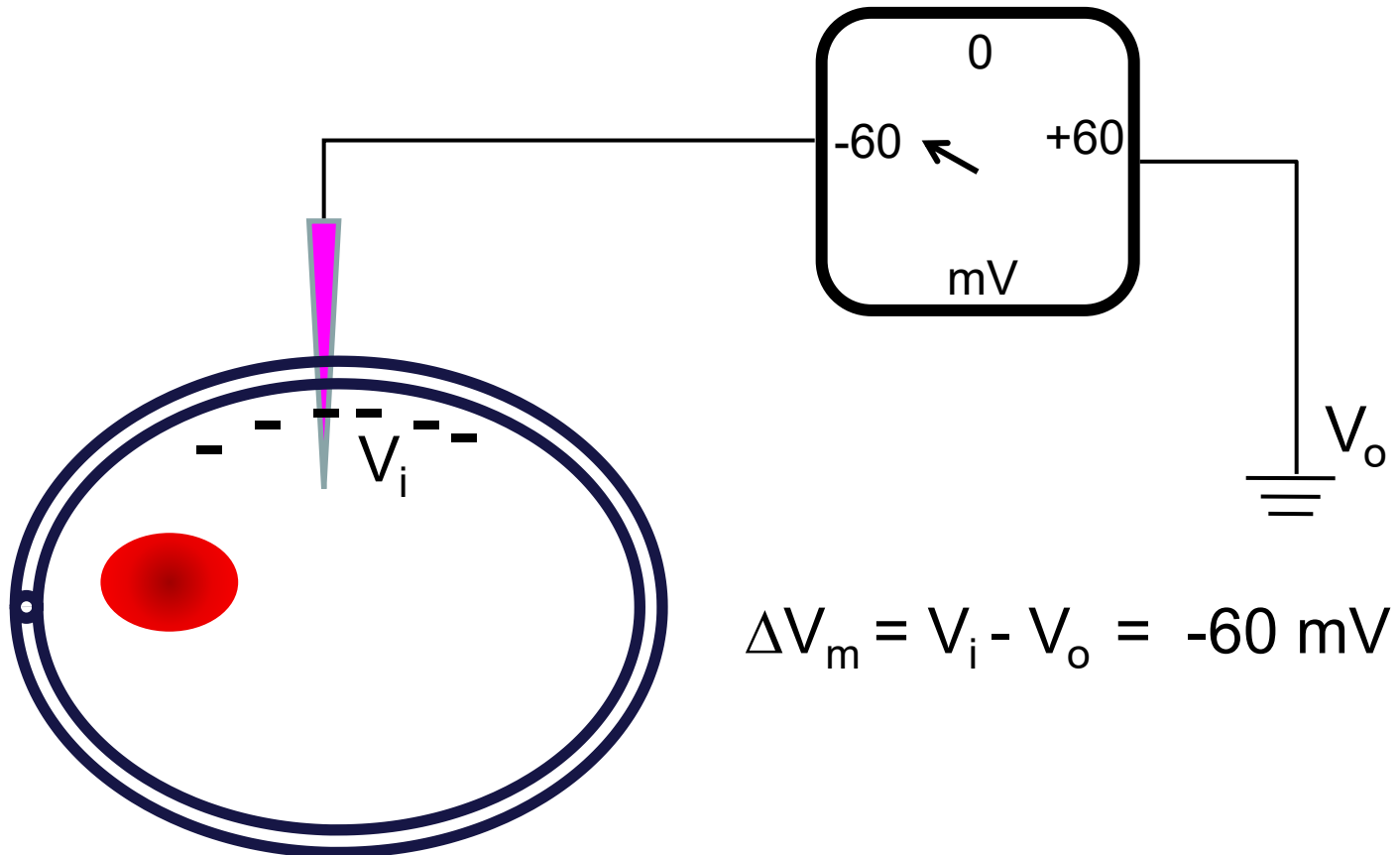
Patch-clamp recordings



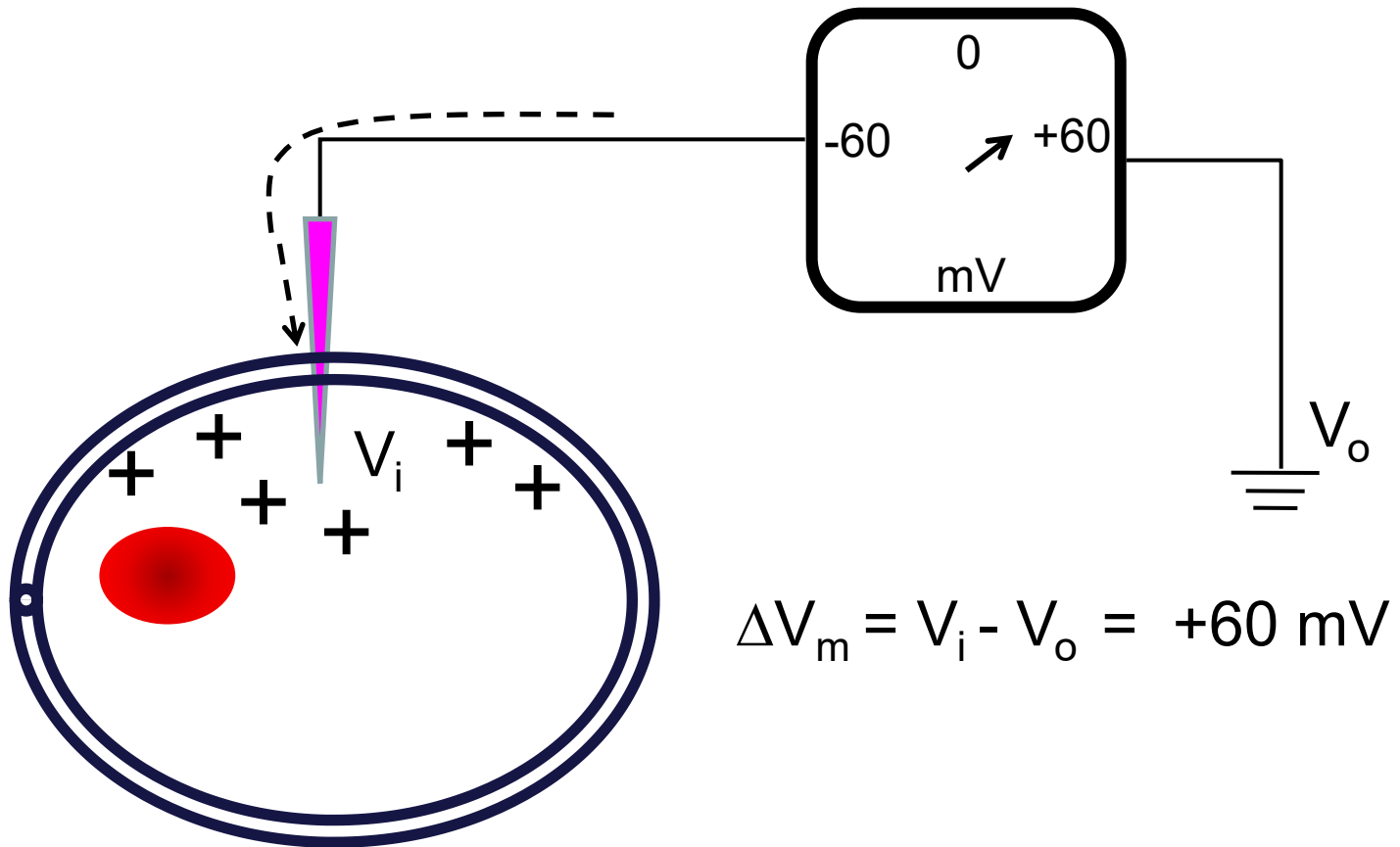
¿How to measure the membrane potential?



¿How to measure the membrane potential?



Intracellular recordings allow for current injections that modify the membrane potential



Patch clamp
Erwin Neher and Bert Sakmann

Nobel Prize in Physiology or Medicine 1991



Photo from the Nobel Foundation archive.

Erwin Neher

Prize share: 1/2



Photo from the Nobel Foundation archive.

Bert Sakmann

Prize share: 1/2

Erwin Neher and Bert Sakmann's patch-clamp technique is based on an exquisitely simple idea. An extremely fine glass pipette with a very small opening is used to make contact with a tiny area, or patch, of the cell's outer membrane that, with some luck, will contain only a single ion channel. Applying a small amount of suction through the pipette forms a seal tight enough such that ions can only flow from the channel into the pipette. By fitting the pipette with a highly sensitive electrode, Neher and Sakmann could record every minute change in current produced as ions flow through the clamped channel into the pipette. With their recording device, they showed how these channels function by opening up and closing in nerve cell membranes to allow certain ions through one at a time.

Patch clamp

Modes:

Voltage clamp (V-C)

Current clamp (I-C)

→ $I=0$ (voltage follower)

Configurations:

Classic:

Cell-attached

Excised inside-out

Whole-cell

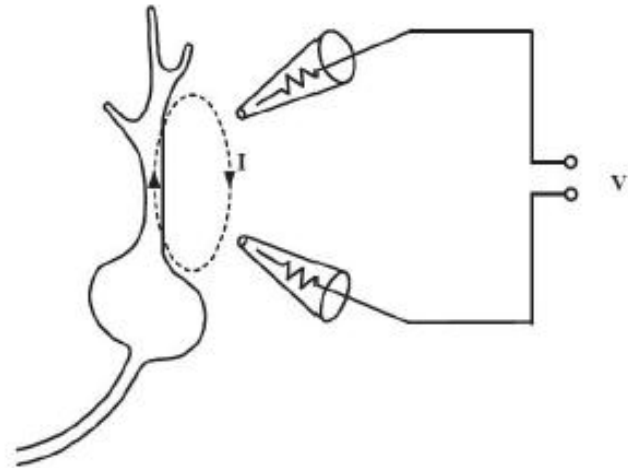
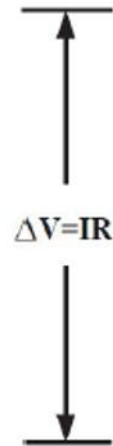
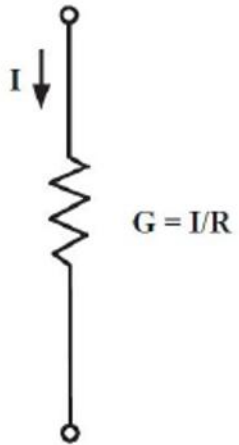
Exquisite:

Loose patch

Perforated patch

Excised Outside-out

Basis: Ohm's law



Basis: The voltage divider

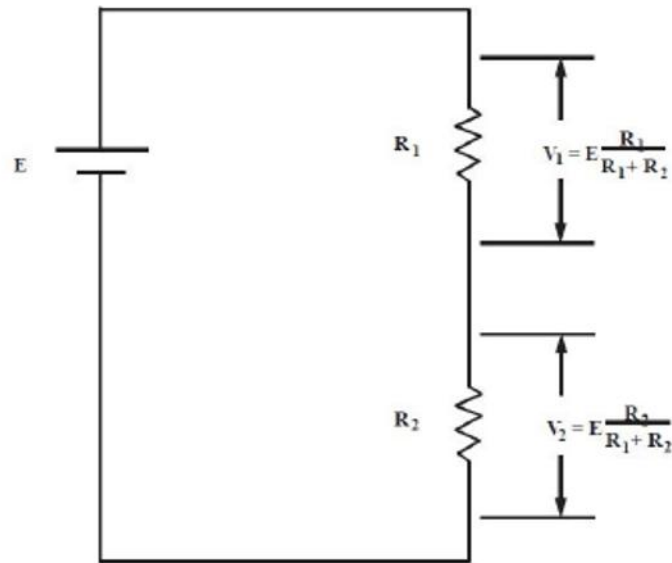
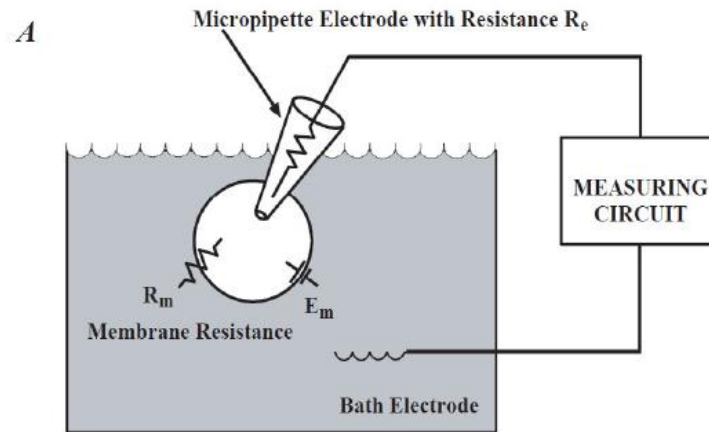


Figure 1-7. A voltage divider.

The total potential difference provided by the battery is E ; a portion of this voltage appears across each resistor:

$$\Delta V_1 = E \frac{R_1}{R_1 + R_2} ; \Delta V_2 = E \frac{R_2}{R_1 + R_2} \quad (2a)$$

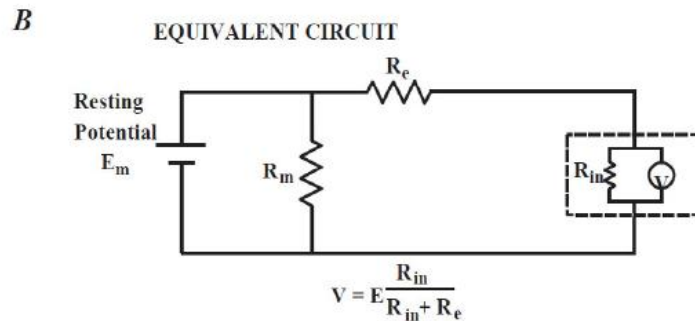
Basis: The voltage divider



R_e - R_{in} forms a voltage divider

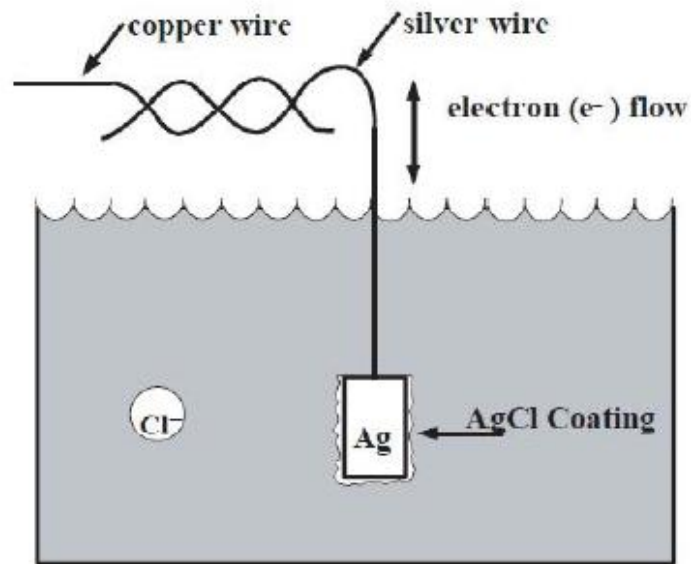
Measured $V = E_m * R_{in} / (R_{in} + R_e)$.

The larger R_{in} , the closer V is to E_m . Best solution: to make R_{in} as large as possible. (in practice 10^{11} - $10^{13} \Omega$)



-Best way to measure current: to open the circuit and insert an ammeter (hopefully with zero resistance)

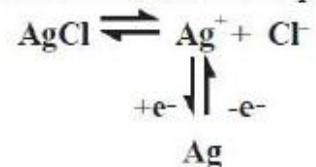
Basis: The silver-chloride electrode

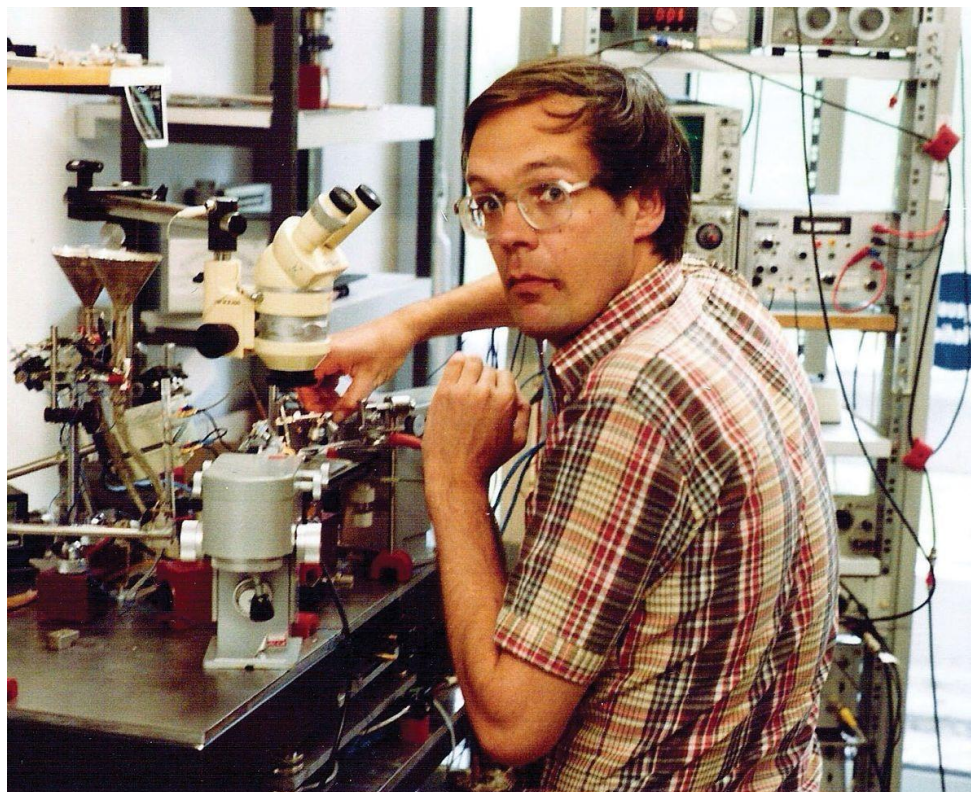


Electrode reaction:



This reaction can also be presented by:

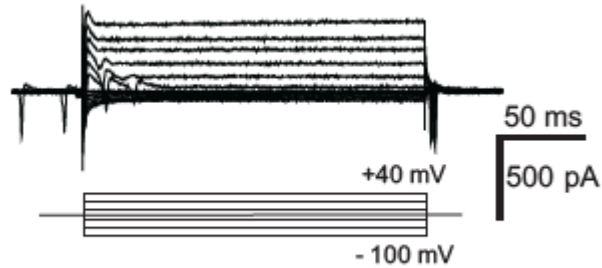




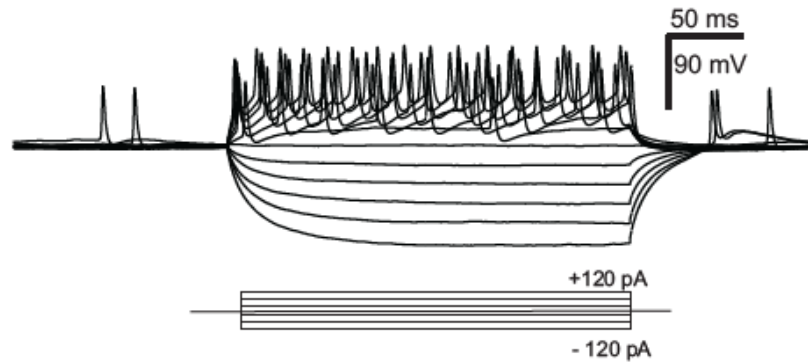
....ready to make some recordings

Patch clamp: modes

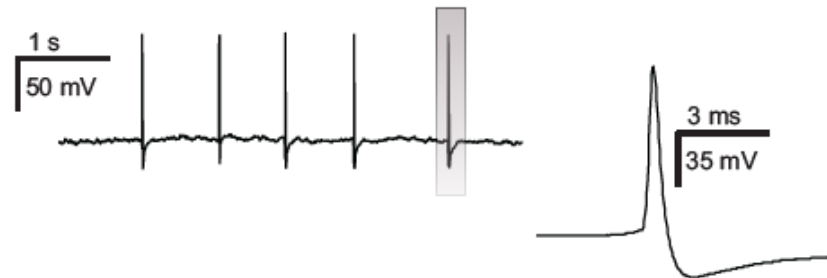
Voltage clamp (V-C)



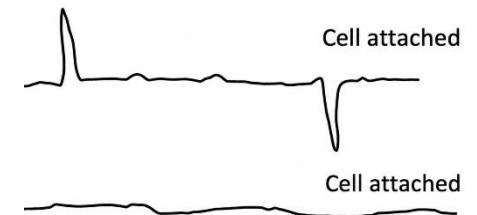
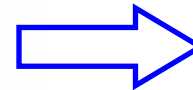
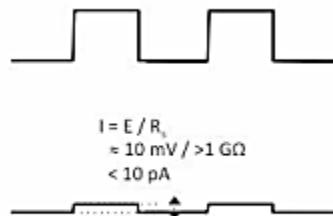
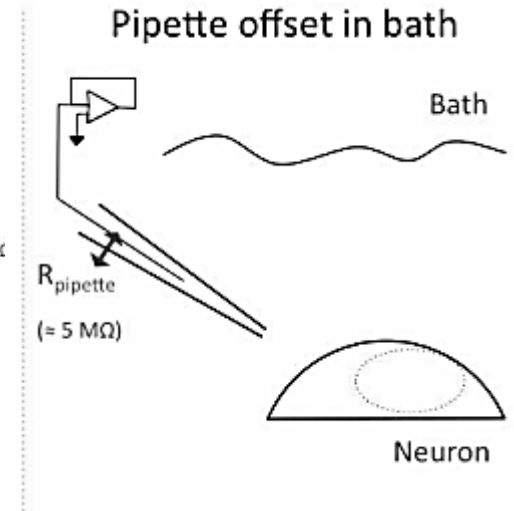
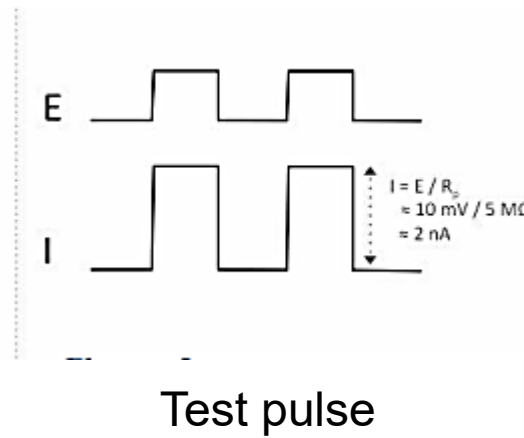
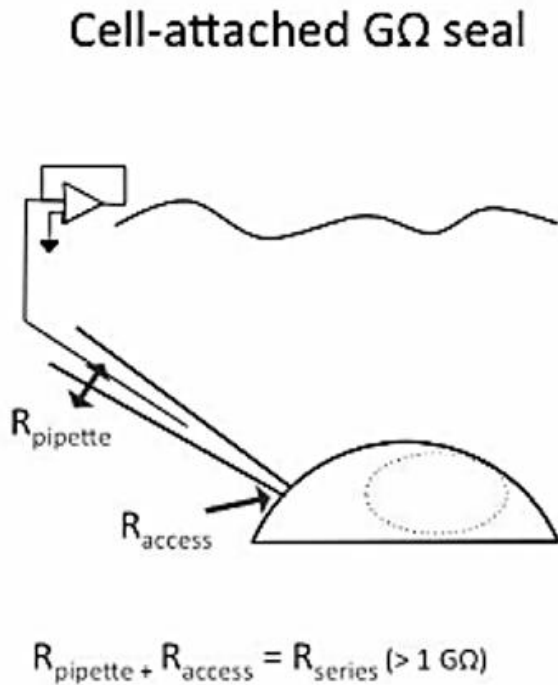
Current clamp (I-C)



→ $I=0$ (voltage follower)

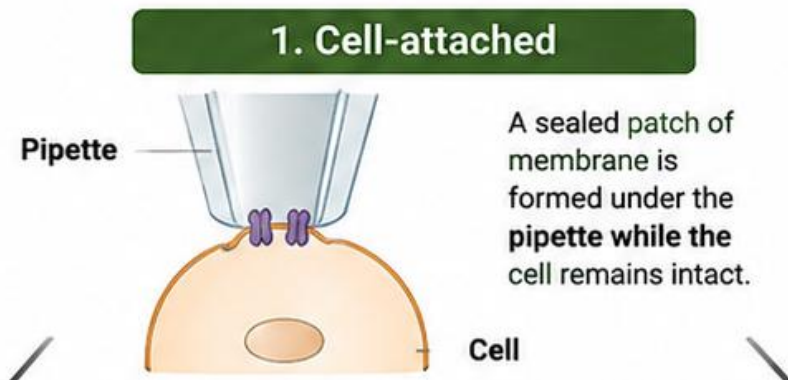


Patch clamp configurations: cell-attached

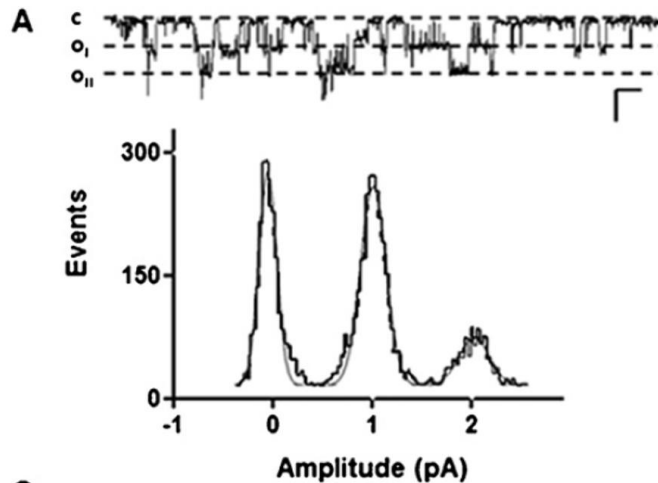


$$I_c = C_m \frac{dV_m}{dt}$$

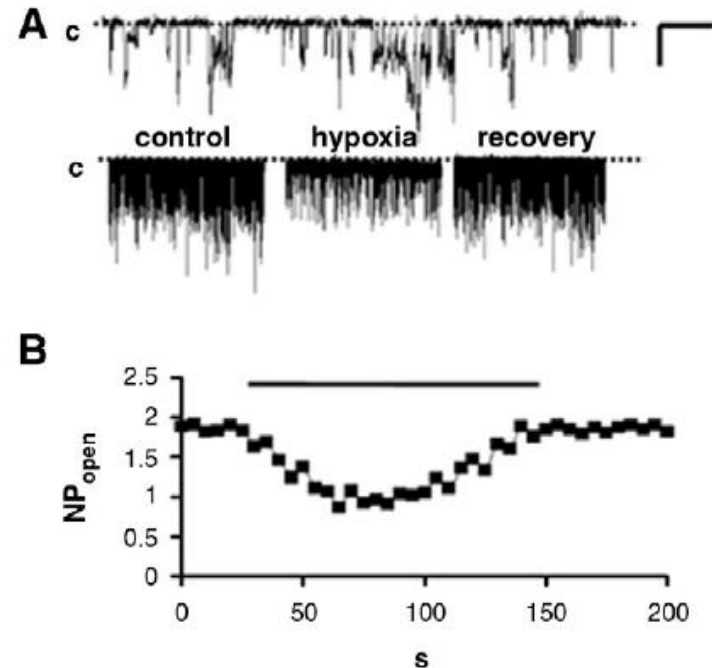
Patch clamp configurations: cell-attached



Allows for
single-or few-channel current recordings
(Applied $V \rightarrow$ outside-to-inside)

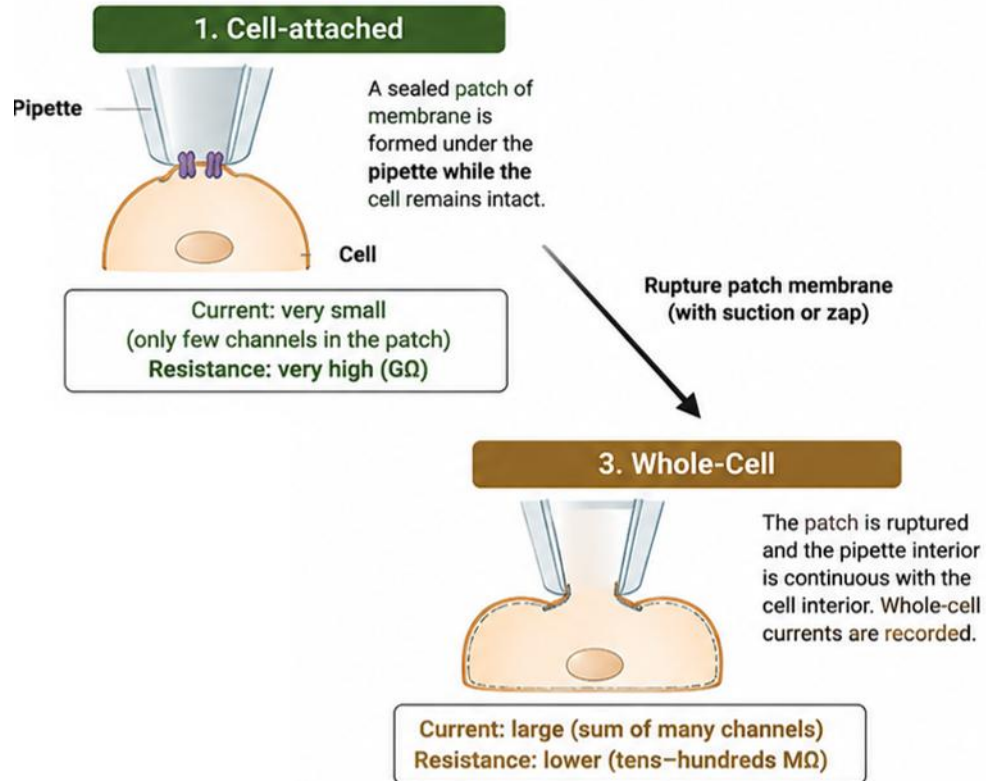


Ortiz et al., 2013



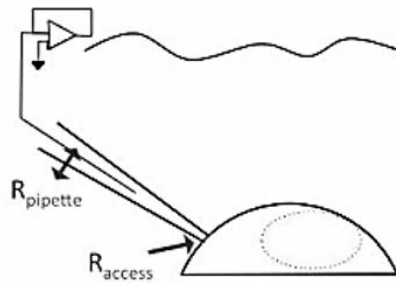
Ortiz and Varas, 2010

Patch clamp configurations: whole-cell

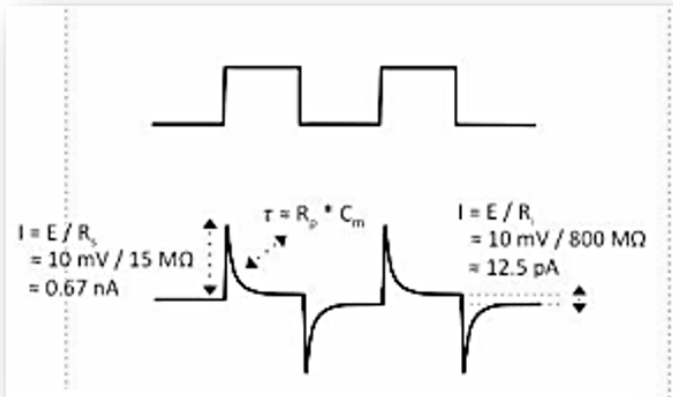


Patch clamp configurations: whole-cell

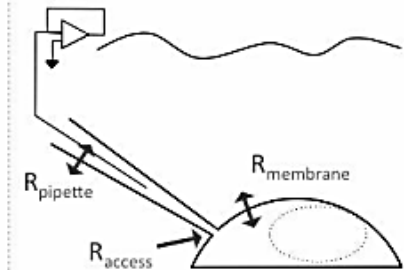
Cell-attached GΩ seal



$$R_{\text{pipette}} + R_{\text{access}} = R_{\text{series}} (> 1 \text{ G}\Omega)$$



Whole-cell patch clamp

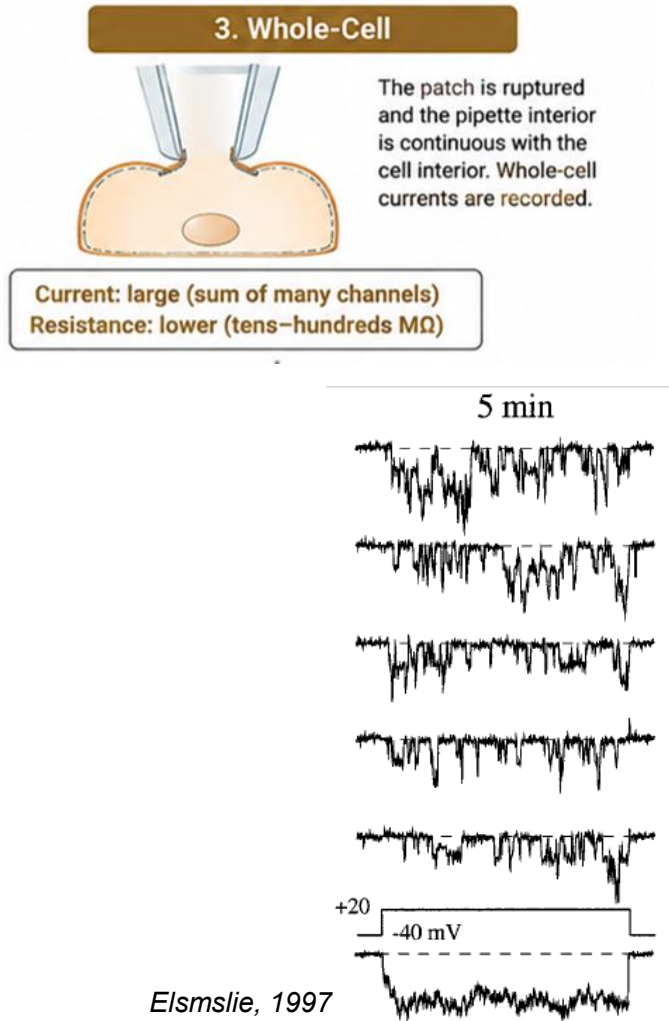


$$R_{\text{pipette}} + R_{\text{access}} = R_{\text{series}} (\approx 15 \text{ M}\Omega)$$

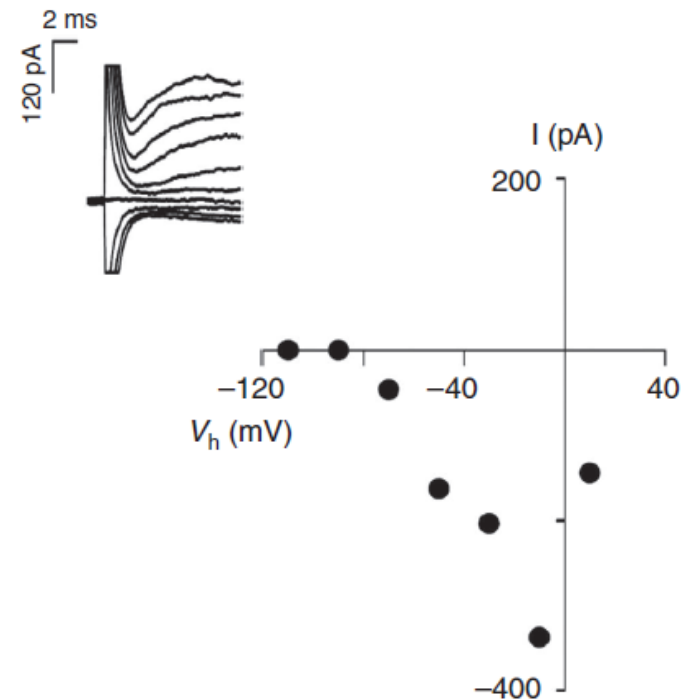
$$R_{\text{series}} + R_{\text{membrane}} = R_{\text{input}} (\approx 800 \text{ M}\Omega)$$



Patch clamp configurations: whole cell

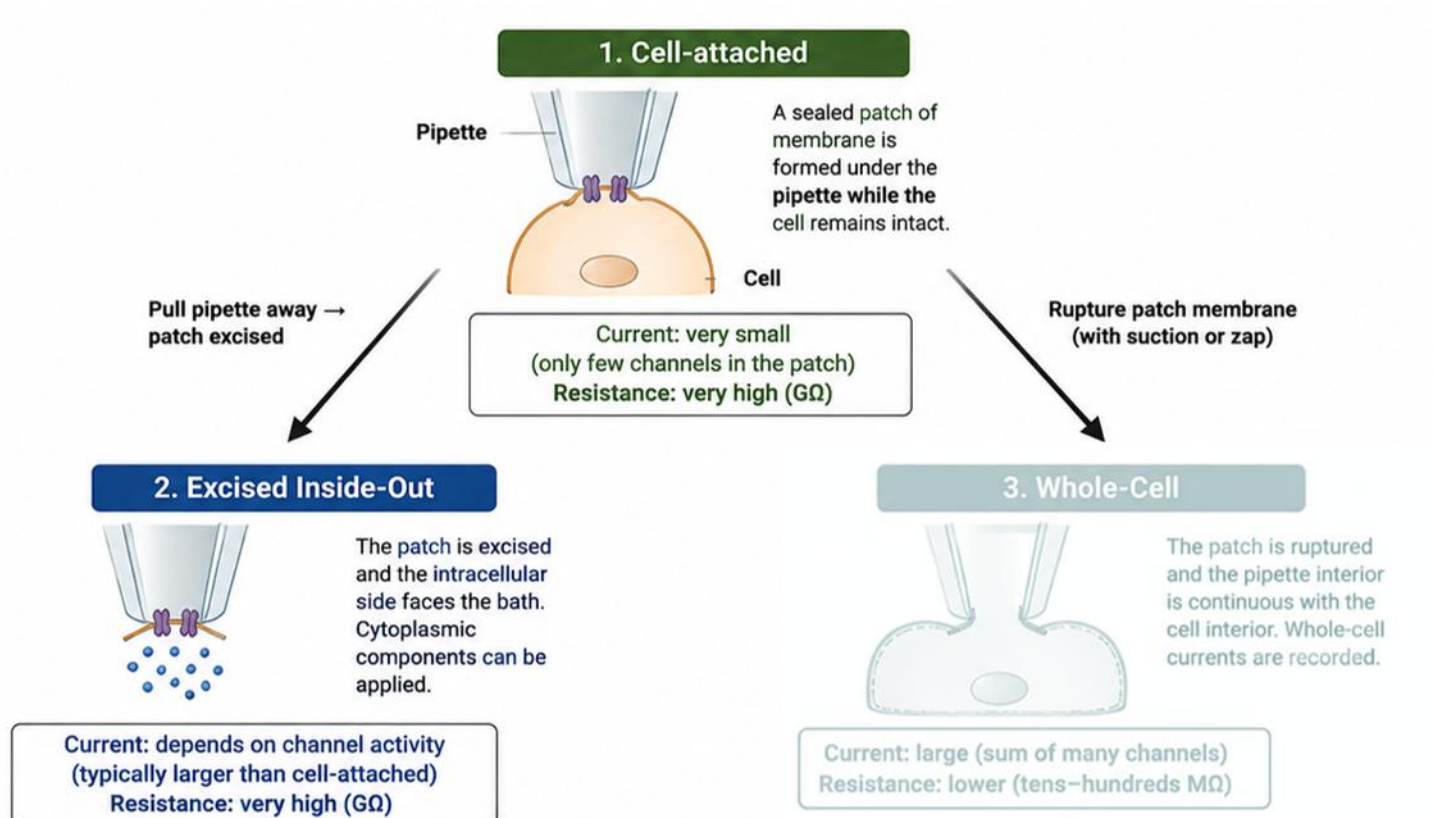


Allows for
Macroscopic current recordings
(Applied $V \rightarrow$ inside)

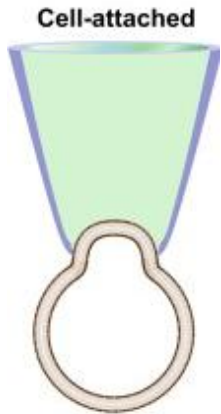


Wake et al., 2015

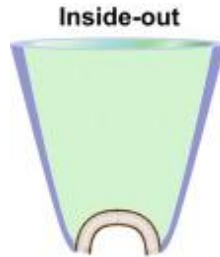
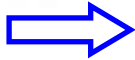
Patch clamp configurations: Excised inside-out



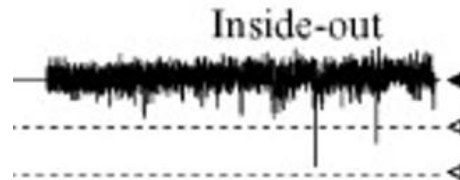
Patch clamp configurations: Excised inside-out



Pulling



Allows for
Controlling intracellular conditions
(Applied $V \rightarrow$ outside-to-reference)



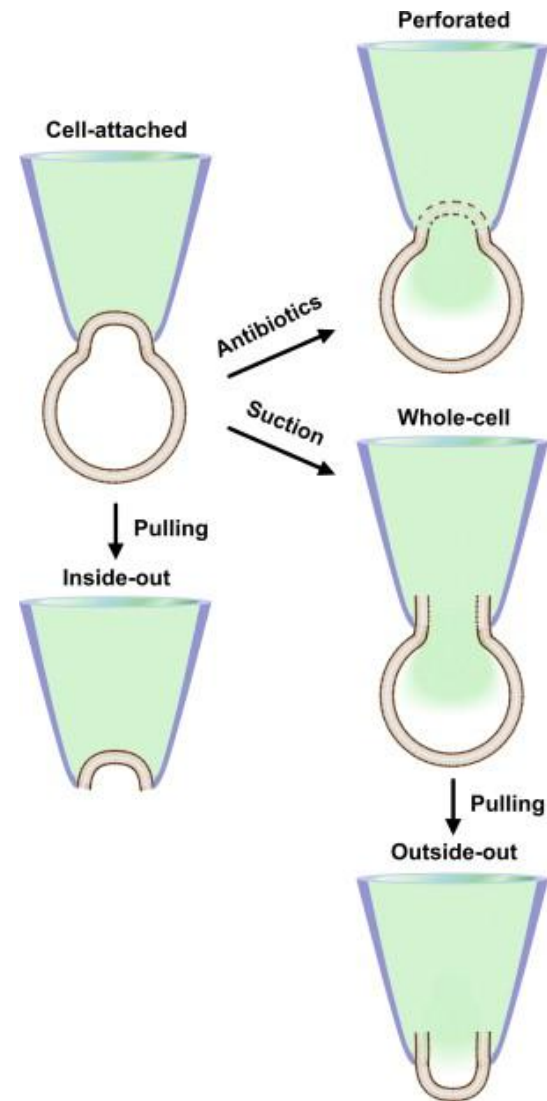
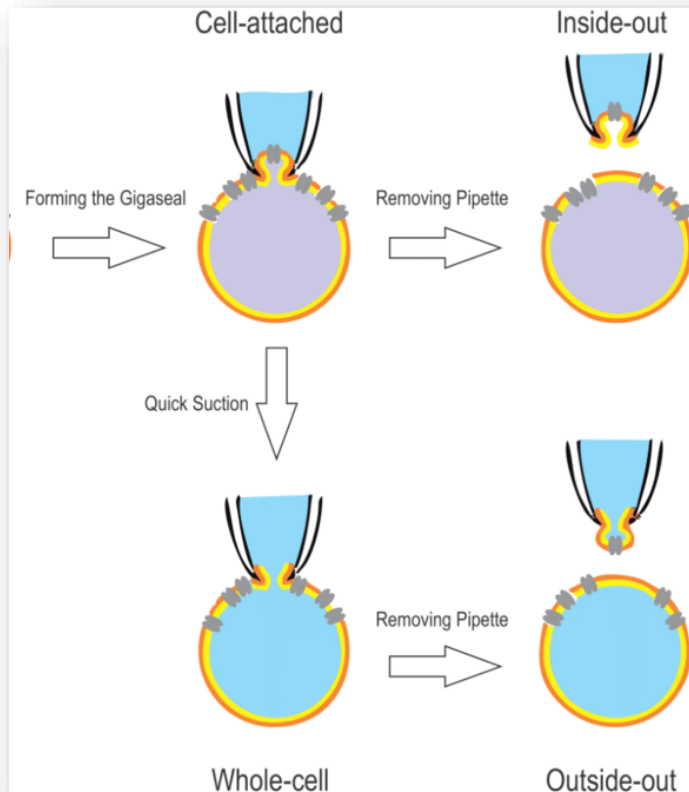
Phosphatase inhibited

Patch clamp configurations: Exquisite level

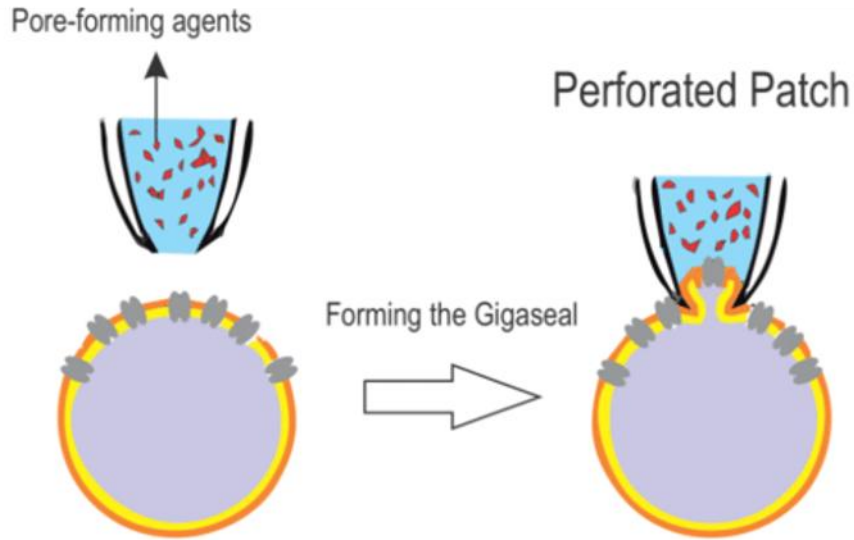
Perforated patch

Excised Outside-out

Loose patch



Patch clamp configurations: perforated patch

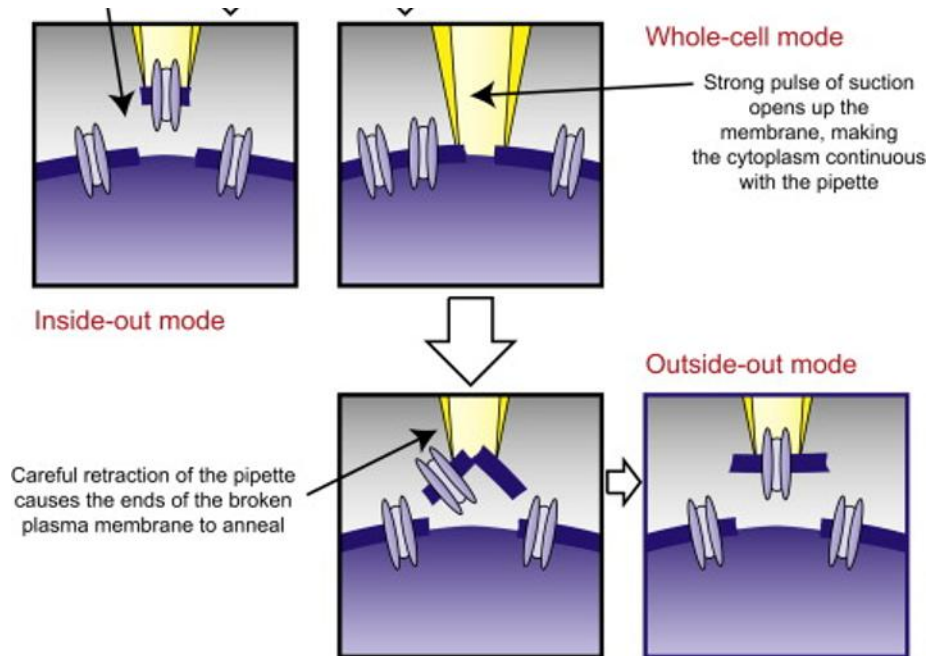


Allows for
Preserving membrane/reduced dialysis
“actual” RMP

Amphotericin B/ Nystatin Forms pores permeable preferentially to small monovalent cations (such as Na^+ , K^+ , but also Cl^-) blocking divalent cations. Pore around 0,8 nm ** it might requires a solvent like DMSO.

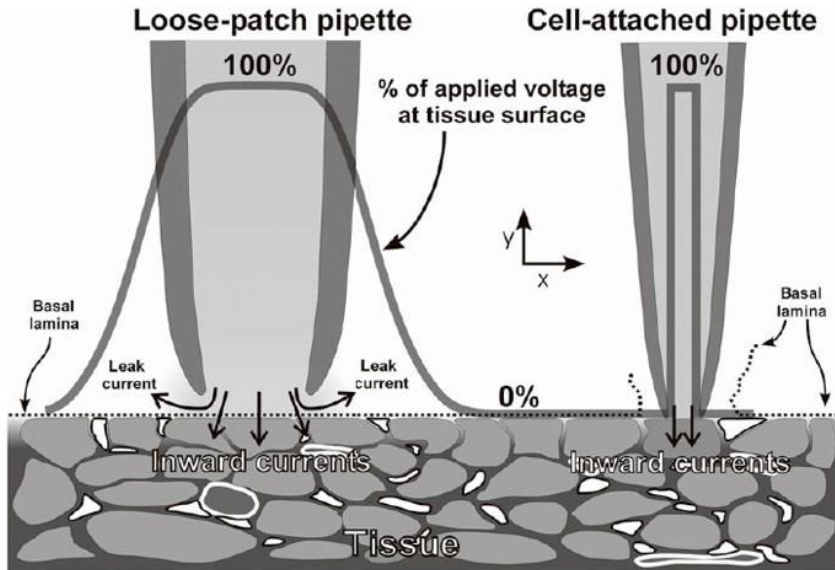
Gramicidin Forms cation-selective pores that are strictly impermeable to anions (like Cl^-) and divalent ions. Pore around 0,4 nm. ** Highly valuable for resting membrane potential measurements

Patch clamp configurations: outside-out

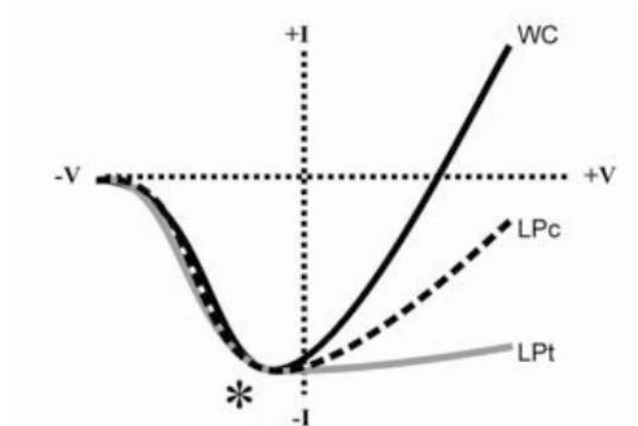


Allows for
Controlling extracellular conditions
(Applied $V \rightarrow$ inside-to-reference)

Patch clamp configurations: loose patch

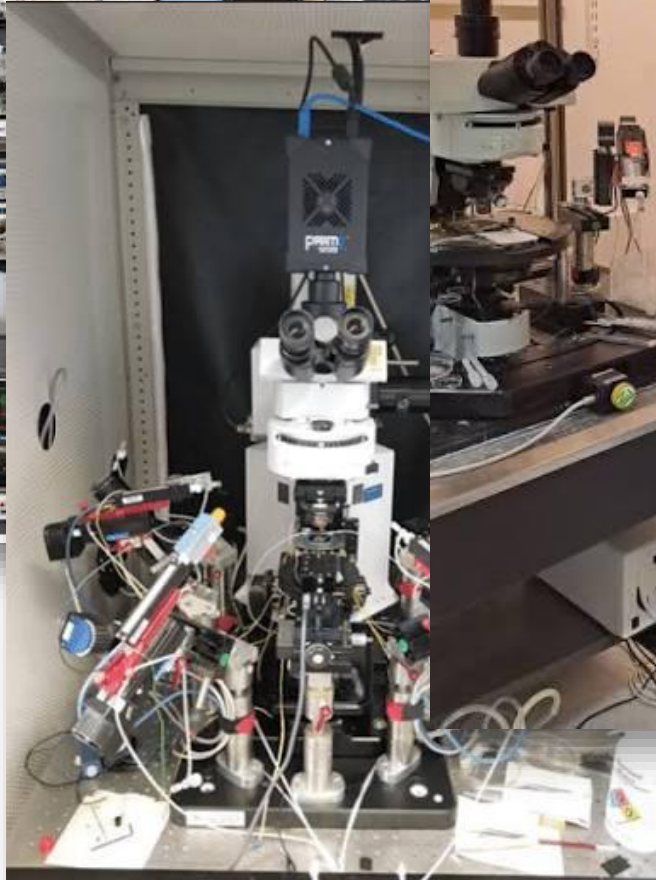


Allows for
Non-invasive recordings (long-lasting)

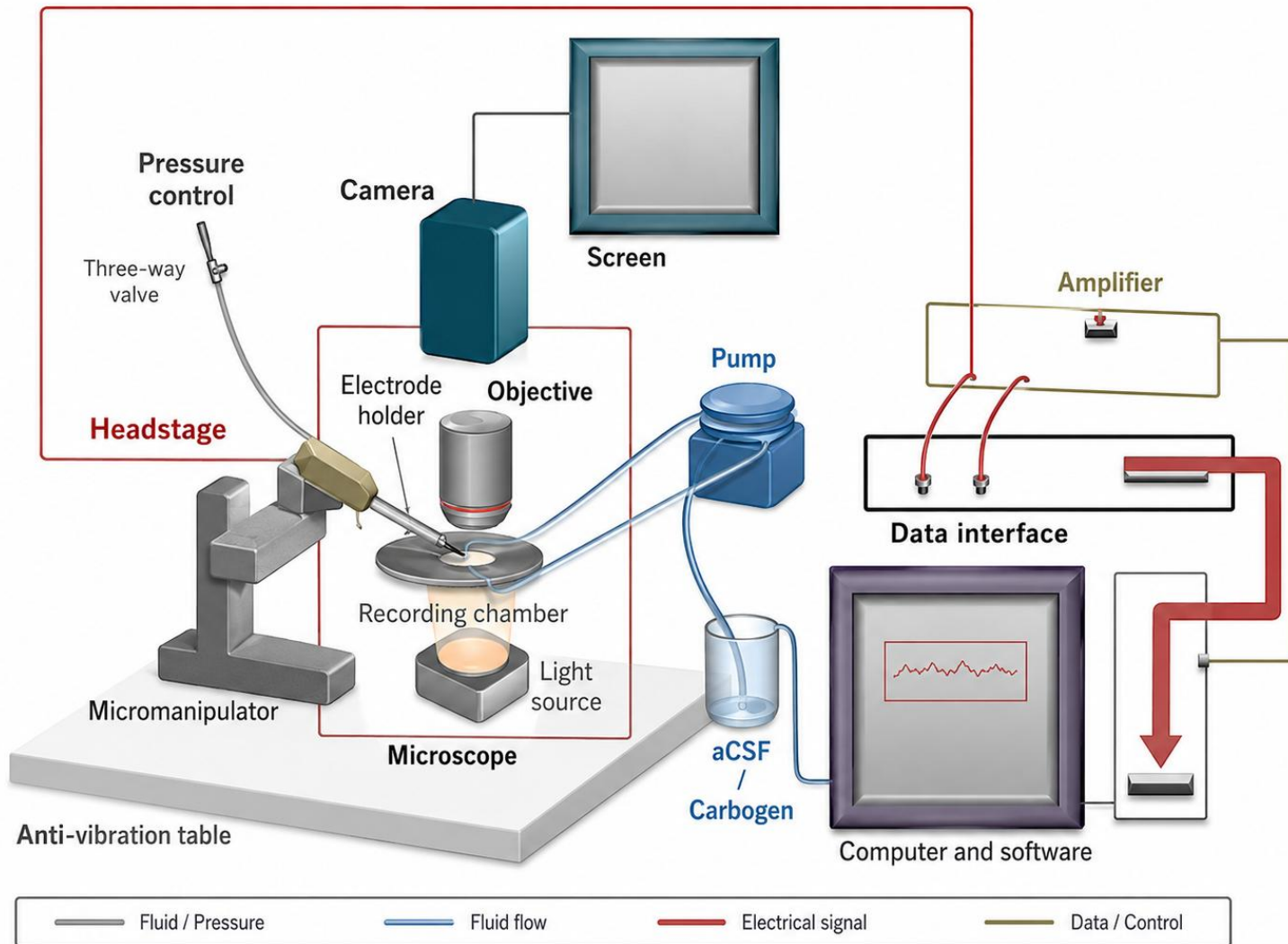


Marrero, H.G., Lemos, J.R. (2007). Loose-Patch-Clamp Method. In: Walz, W. (eds) Patch-Clamp Analysis. Neuromethods, vol 38. Humana Press. https://doi.org/10.1007/978-1-59745-492-6_11

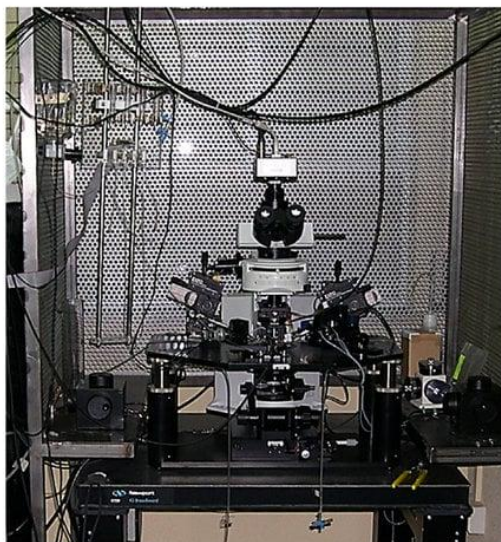
real life....







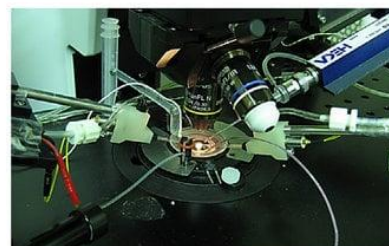
A



B



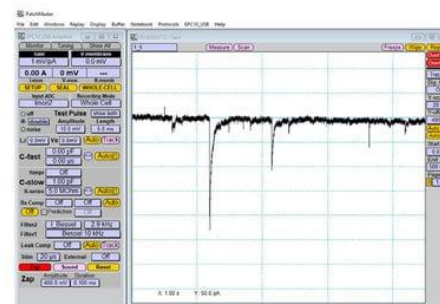
C



D



E



The Patch Clamp Method

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THE PATCH CLAMP TECHNIQUE

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Universidad de Santiago de Chile

Nernst Potential

$$\Delta G_{\psi} = \Delta G_c$$

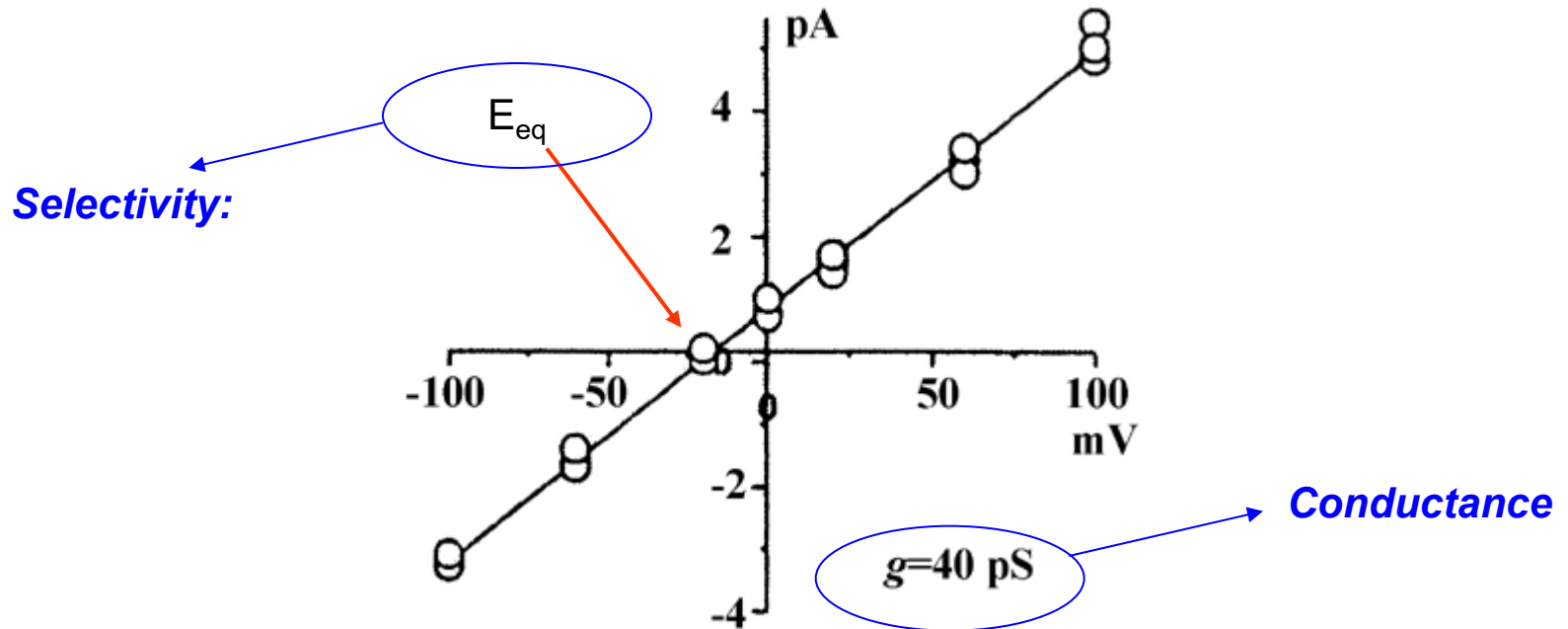
$$\Delta G_{\psi} = -zF\Delta V$$

$$\Delta G_c = -RT \ln (C_e/C_i)$$

$$zF\Delta V = RT \ln (C_e/C_i)$$

$$\Delta V = \frac{RT}{zF} \ln (C_e/C_i)$$

Nernst Potential



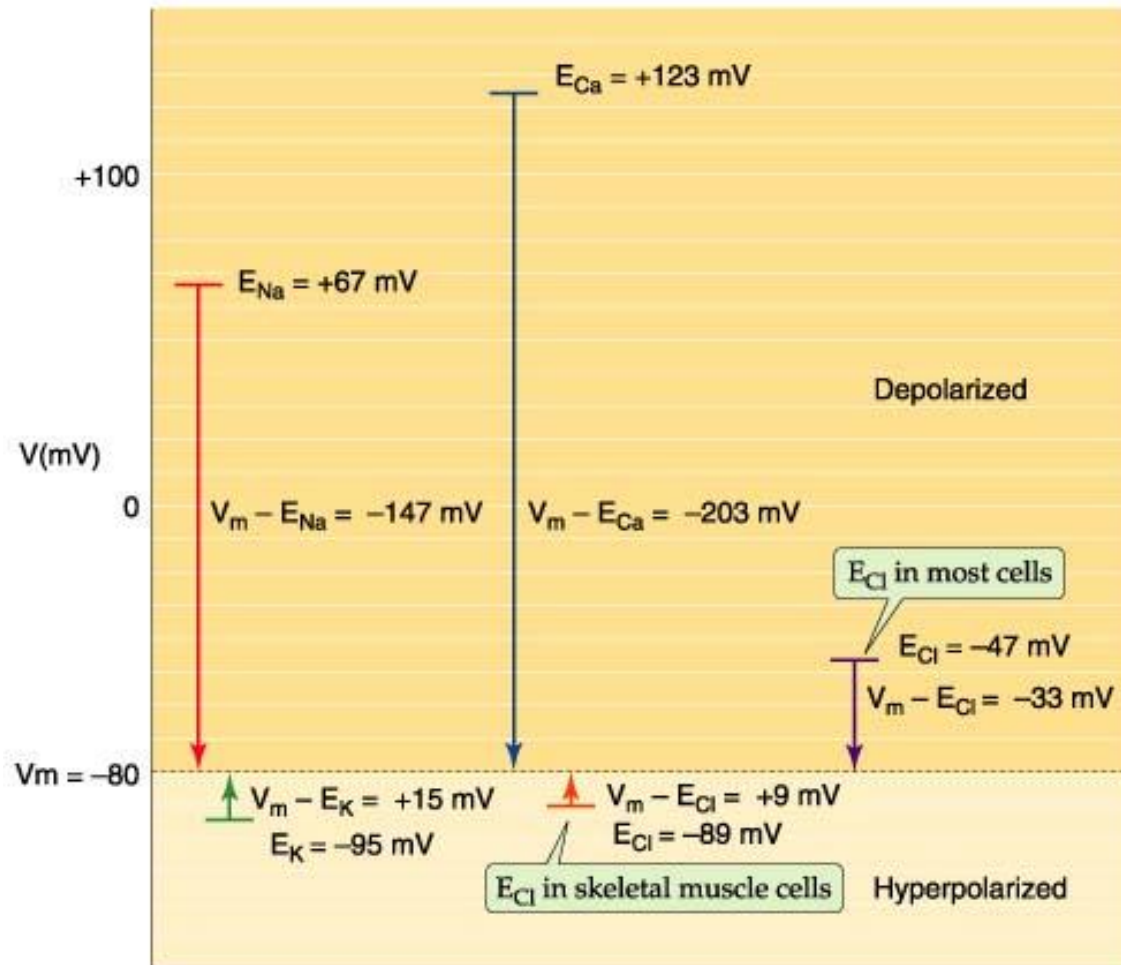
$$\Delta V = I R$$

What is the driving force?

$$(V_m - E_{eq})$$

Nernst Potential

... determines both transport *driving force and direction* ions.



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$$(V_m - E_{eq})$$

El potencial de membrana en reposo:

Ecuación de Nernst:

$$E_K = \frac{RT}{zF} \ln \frac{[K]_o}{[K]_i}$$

- Relaciona la gradiente de concentración y la gradiente eléctrica para un ion determinado.
- Permite predecir el movimiento del ión en cuestión.
- Considera sólo iones aislados.

Ecuación de GHK:

$$V_m = \frac{RT}{zF} \ln \frac{P_K [K]_o + P_{Na} [Na]_o + P_{Cl} [Cl]_i}{(P_K [K]_i + P_{Na} [Na]_i + P_{Cl} [Cl]_o)}$$

- Relaciona la gradiente de concentración y la gradiente eléctrica para todos los iones presentes.
- Permite predecir el valor del potencial de membrana.
- Cuantifica la contribución de diversas especies iónicas.

$$\text{Fuerza Impulsora} = (V_m - E_{eq})$$

